

Mid Semester Examination

Session 2025-2026 (Winter Semester)

Subject: Thermodynamics (NMEC103)

Time: 60 minutes

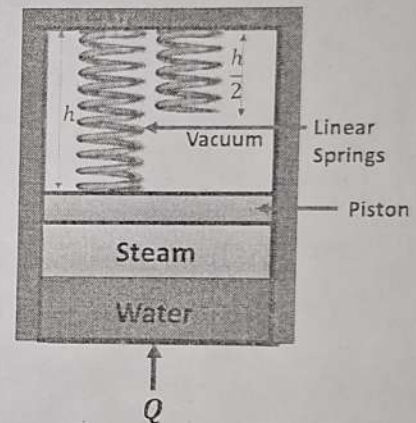
Max. Marks: 60

Answer all the questions. Symbols used have their usual meaning. No steam table is allowed.

Assume missing data suitably, if any.

- 1
- (a) What does it mean by a **Simple Compressible Pure substance**? For non-simple, pure substances, list all the forms of energy that the stored energy **E** contains. Clearly state which component accounts for thermal energy and what this thermal energy consists of. Is this **E** a low-grade or high-grade energy? If a solid bar is distorted within its elastic limit, the associated strain energy will be stored in which component of **E**? Is the flow energy **pv** a part of **E**? [8 Marks]

- (b) In the system shown in the figure, both liquid water and steam are in equilibrium. What is the state of the fluid? The fluid is slowly heated, and it is observed that the vapour mass increases while the liquid mass decreases, and finally everything becomes vapour just at the instant when both springs are fully compressed. Beyond this point, the heating continues. During the phase change process, will the temperature remain constant? If the heating is done at a constant rate, when will the boiling rate be high, at the beginning or towards the end of vaporization? Justify your answer. Show the process on p-v and T-v plots. [8 Marks]

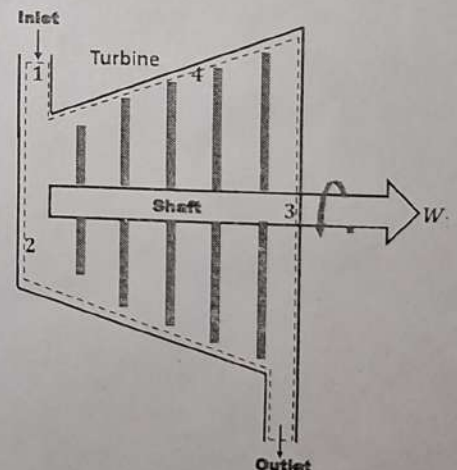


- (c) The 1st law for a control volume can be written as:

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W} + \dot{m}_i \left(u_i + \frac{1}{2} V_i^2 + gZ_i \right) - \dot{m}_e \left(u_e + \frac{1}{2} V_e^2 + gZ_e \right)$$

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W}_{cv} + \dot{m}_i \left(h_i + \frac{1}{2} V_i^2 + gZ_i \right) - \dot{m}_e \left(h_e + \frac{1}{2} V_e^2 + gZ_e \right)$$

What is the difference between the $\dot{W}$ and the $\dot{W}_{cv}$. What does this $\dot{W}_{cv}$ include in general. Does it include the Pdv work? When the working substance flows through a control volume at the control surface, both normal and shear forces act. Consider a turbine shown in the figure, which has an output shaft, a rigid boundary, and flow inlets and exits. Comment on the work associated with shear and normal forces at the different points shown in the figure. Assume the flow velocity to be normal to the cross-sections at the inlet and exits. In other places, the turbine wall is impermeable. [7 Marks]



- (d) A Pressure cooker has the lid screwed on tight. A small opening with an area $A = 5 \text{ mm}^2$ is covered by a petcock that can be lifted to allow steam to escape. How much mass should the petcock have to allow boiling at 120°C with an outside atmospheric pressure of 101.3 kPa ? How much less energy (per kg of water) is required to boil in this pressure cooker than in an open container? [7 Marks]

Temperature, $^\circ\text{C}$	Saturation Pressure (kPa)	v_f (m^3/kg)	v_g (m^3/kg)	u_f (kJ/kg)	u_g (kJ/kg)
120	198.5	0.001060	0.89186	503.48	2529.24
100	101.3	0.001044	1.67290	418.91	2506.50

2. NASA astronauts Sunita Williams and Butch Wilmore came back from the International Space Station (ISS) concluding a **286-day** mission aboard a SpaceX Crew Dragon reentry capsule (named Freedom), which reentered Earth's atmosphere and splashed down in the ocean. When reentry capsule is falling through the earth's atmosphere, its motion is being resisted by the aerodynamic forces, while the friction between air and the capsule surface is causing its temperature to increase.

The surface of the capsule is not perfectly insulated, resulting in undesirable *Heat Transfer* between it and the surrounding atmosphere.

- (a) Does the reentry capsule define a closed system or an open system?
- (b) Taking into account its velocity v , acceleration a , altitude H , average temperature T , aerodynamic drag force F_v , derive the Energy Balance Equation following First Law of Thermodynamics. Assume other quantities as you deem appropriate. [3+12=15 Marks]

3. A thermoelectric generator consists of a series of semiconductor elements heated on one side and cooled on the other. It is a type of thermal engine, except that the output is electrical rather than mechanical work. Electric direct current output is produced as a result of an input heat transport of energy. In a particular experiment, the steady state direct current is measured to be **0.500 Ampere** and the potential across the unit is **0.800 Volts**. The heat input to the hot side is **5.50 Watt**. Determine the heat transfer rate from the cold side and the energy conversion efficiency of this device. [15 Marks]
